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United States Patent Application Publication

20250280808

Kind Code

A1

Publication Date

September 11, 2025

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Spring Actuated Weedless Fishing Lure

Abstract

A spring actuated weedless fishing lure increases the chances of successfully catching a fish without entangling a hook on plant material. A weedless guard system prevents the hook from catching on weeds and compresses a spring to hold the spring and hook in a retracted position. When a fish strikes, the weedless guard system is released allowing the spring to expand and push the hook in a longitudinal direction to an expanded position away from other parts of the lure. This expanded position provides a greater area for the fish's mouth to engage with the hook without interference from other parts of the lure. The expanded position also allows the hooks to embed closer to the front of a fish's mouth and reduces the risk of the fish swallowing the whole lure.

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Family ID: 1000008620999

Appl. No.: 19/217222

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18142705 20230503 parent-grant-document US 12336518 child US 19217222

parent US continuation 16008139 20180614 parent-grant-document US 11684053 child US 18142705

us-provisional-application US 62520299 20170615

Publication Classification

Int. Cl.: A01K85/02 (20060101); A01K83/02 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Nonprovisional application Ser. No. 18/142,705 filed on May 3, 2023, which is a continuation of U.S. Nonprovisional application Ser. No. 16/008,139 filed on Jun. 14, 2018, now U.S. Pat. No. 11,684,053 B2, which claims the benefit of U.S. Provisional Application Ser. No. 62/520,299 filed on Jun. 15, 2017.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] This invention relates to a spring actuated weedless fishing lure that projects one or more hooks in a longitudinal direction to more securely attach to a fish upon engaging the lure.

2. Description of Related Art

[0003] Weedless fishing lures provide a mechanism to prevent the hook from snagging on plant material, such as a tree limb or lily pad, while fishing. These types of lures are well known in the prior art. For example, U.S. Pat. No. 3,670,446 discloses a single or double hook lure with a guard that holds the end of the hook(s) so they do not hook into any plant material. Once engaged by a fish, the hook end is released from the guard allowing the hook to embed in the fish's mouth. In another example, U.S. Pat. No. 5,628,138, the hook end is hidden within a cavity of the lure body (shaped like a small fish for example). The hook is attached to a “spring member” that has an upper, stiff curved portion and a lower flexible portion of spring steel or resilient plastic. When engaged by a fish, the lower flexible portion is pushed toward the stiff portion, which causes the hook to move in an arcuate direction out of the cavity to embed in the fish's mouth.

[0004] These weedless prior art lures are effective at preventing the hook from getting caught on plant material; however, the guards that prevent the hook from snagging on plants and tree limbs can also result in missing a fish altogether. Once a fish strikes the lure, the guard mechanisms or other aspects of the lure body can interfere with the hook or provide only a small area for the fish's mouth to engage with the hook making it difficult for the hook to sufficiently embed in the mouth to catch the fish. Another problem with prior art lures, particularly in bass fishing, is that the fish can end up swallowing the whole lure. If there is a need to release the fish, it can be impossible to remove a swallowed lure without significantly harming the fish. There is a need for a weedless hooking system that does not interfere with embedding the hook in the fish's mouth, reduces the rate of catch failure after a strike, and reduces the risk of the fish swallowing the lure.

SUMMARY OF THE INVENTION

[0005] According to one preferred embodiment, a spring actuated weedless lure according to the invention comprises a housing, a spring disposed within the housing and configured to move a hook in a substantially longitudinal direction (relative to the housing), and a guard connected to the housing and releasably connected to the hook. When connected to the hook, the guard holds the hook and spring in a retracted position. Once a fish strikes, the connection between the guard and hook is released, allowing the spring to expand and to launch or move the hook from a retracted position to an extended position along a longitudinal axis relative to the housing. The hook is preferably extended out 1-3 inches from the proximal end of the housing when in an extended position. This allows the lure to have the benefits of a weedless hook but also prevents the guard or housing from interfering with catching the fish. The extended hook position also increases the chances of catching a fish since they are positioned in a larger exposed area with more distance

between the hook and the rest of the lure, providing a greater chance of being embedded in the fish's mouth. It also allows the hook to embed in the first inch or two of the fish's mouth, without allowing the entire lure to be swallowed, making it easier to release the fish if desired.

[0006] According to another preferred embodiment, a spring actuated weedless lure like that described above is contained within a hollow body, preferably shaped like a frog, worm, small fish, or other design that would attract a fish to strike the lure. The housing, guard and hook are preferably disposed entirely within the hollow body when the spring is in a retracted position. The hollow body preferably has one or more apertures to allow the hook or hooks to extend out away from the body when the spring is released and the hooks or hooks are in an extended position. Most preferably, the hook or hooks are extended 1-3 inches beyond the proximal end of the hollow body when in an extended position. When a hollow body is used, the guard need not engage with the barb end of the hook to provide a weedless design because the hooks are disposed inside and protected from entanglement with plant material by the hollow body. A guard that does engage the barb ends of the hook, similar to that disclosed in U.S. Pat. No. 3,670,446 or in other prior art, may also be used with various embodiments of the invention with or without a hollow body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The system and method of the invention is further described and explained in relation to the following drawings wherein:

[0008] FIG. 1 is a top plan view of a double hook fishing lure in a retracted position according to one preferred embodiment;

[0009] FIG. 2 is a side elevation cross sectional view of the fishing lure of FIG. 1;

[0010] FIG. 3 is a perspective cross sectional view of the fishing lure of FIGS. 1 and 2; and

[0011] FIG. 4 is side elevation cross sectional view of the fishing lure of FIG. 1 in an extended position;

[0012] FIG. 5 is a perspective cross sectional view of the fishing lure of FIG. 4;

[0013] FIG. 6 is a side elevation cross sectional view of the fishing lure of FIG. 1 within one embodiment of a decorative hollow body;

[0014] FIG. 7 is an end elevation of the embodiment shown in FIG. 6;

[0015] FIG. 8 is a perspective cross sectional view of another preferred embodiment of a fishing lure in a retracted position; and

[0016] FIG. 9 is a perspective cross sectional view of the embodiment of FIG. 8 in an extended position.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0017] Referring to FIGS. 1-5, one preferred embodiment of a spring actuated weedless lure 10 is depicted. Lure 10 preferably comprises a housing 12, an attachment mechanism 14, a weedless guard system 16, and a spring 28 disposed inside housing 12. Attachment mechanism 14 is preferably a ring or eye bolt secured to one end of housing 12 to allow lure 10 to be attached to a fishing line or connected to a hollow body (such as a decorative frog). Weedless guard system 16 preferably comprises a hook 18 and a guard 20. Guard 20 is preferably an elongated, flexible pin or wire loop. One end of guard 20 is connected to housing 12 and the other end of guard 20 is releasably connected to hook 18. Any prior art mechanism for releasably connecting a guard to a hook may be used with guard 20 and hook 18 in weedless guard system 16. When engaged, weedless guard system 16 holds hook 18 in a retracted position, compressing spring 28. When disengaged, weedless guard system 16 releases hook 18 allowing spring 28 to expand and move hook 18 in a substantially linear or longitudinal direction (aligned with or parallel to a longitudinal axis of housing 12) from its retracted position to an expanded position. Weedless guard system 16

is disengaged when external pressure is exerted on guard **20** in a direction substantially perpendicular to the direction of movement of hook **18** when released. When a fish strikes, its mouth exerts pressure in a substantially lateral direction (relative to housing), disengaging weedless guard system **16** and allowing hook **18** to move in a longitudinal direction.

[0018] Hook **18** may be a single hook, but is more preferably a double hook. Each hook **18** preferably comprises a longitudinal body or shaft **30**, an curved or arcuate portion **32**, and a barbed end **34**. When a double hook **18** is used, a single shaft **30** may split or fork prior to arcuate portions **32** to form a double hook or two separate shafts **30** may be used. When a double hook **18** is used, wireless guard system **16** also preferably comprises brace **22** and a barb **24**. Brace **22** is disposed between the two hooks **18** and barb **24** extends from brace **22**. In this preferred embodiment, guard **20** is a single pin or piece of wire with one end connected to housing **12** and the distal end may have an aperture to engage with barb **24**. Alternatively, the distal end of guard **20** may have a small hook or barb to engage with brace **22** (in which case it is not necessary to use barb **24**) or a loop may dangle from brace **22** (in place of barb **24**) to engage with the hook or barb on the end of guard **20**. In another preferred embodiment as shown in FIGS. **8-9**, guard **20** is in a loop configuration with both ends connected to housing **12** and the distal end of the loop engaging with barb **24** to secure hooks **18** in a retracted position. Barb **24** and guard **20** are made of materials that are preferably strong enough to remain engaged during use and flexible enough to allow them to disengage when the force from a fish biting the lure is applied. As additional alternatives, guard **20** may comprise a loop, or a single pin or piece of wire with a distal end having an aperture that is configured to engage with a barbed end of one or both hooks **18** (in which case, brace **22** and barb **24** are not necessary) or may comprise a single pin or piece of wire with a barb at its distal end configured to engage with brace **22** (in which case barb **24** is not needed).

[0019] Housing **12** is preferably substantially cylindrical with one sealed end near attachment mechanism **14**. The other end is at least partially open to allow movement of an elongated shaft **30** or longitudinal portion of hook **18** along a longitudinal axis of housing **12**. Spring **28** is preferably disposed inside housing **12**, between the sealed end of housing **12** and plate **26**, which is also disposed inside housing **12**. Plate **26** is connected or connectable to the elongated shaft portion **30** of hook **18**. Hook **18** and plate **26** may be integrally formed or connected together by adhesive, welding, threaded attachment, or the like. A seal, such as an o-ring, may optionally be placed around plate **26** to provide a water-tight or substantially water-tight seal to prevent water from entering the portion of housing **12** in which spring **28** is disposed. Alternatively, plate **26** may optionally be made of material, such as rubber, that acts as a seal against interior wall surfaces of housing **12** to prevent water from entering the portion of housing **12** in which spring **28** is disposed. This may allow the interior spring cavity of housing **12** to be filled with air or another gas to aid in lure **10** floating at or near the surface of the water, if desired.

[0020] When guard **20** is connected to hook **18** (or barb **24**), it holds hook **18** and plate **26** in a retracted position, compressing spring **28** inside housing **12** as shown in FIGS. **1-3**. When a fish strikes, its mouth puts pressure on guard **20**, releasing it from hook **18** (or barb **24**). When guard **20** is disengaged from hook **18** (or barb **24**), plate **26** is no longer held in a retracted position allowing spring **28** to expand and push plate **26** rearwardly inside housing **12**. This movement of plate **26** also causes hook **18** to move rearwardly, extending it along a longitudinal axis relative to housing **12** to an extended position as shown in FIGS. **4-5**. The direction of hook movement is substantially longitudinal and not arcuate as in some prior art lures. Most preferably, the extended position for hook **18** is around 1-3 inches from the retracted position of hook **18** or around 1-3 inches the proximal end of a decorative hollow body (if used). The partially open end of housing **12** may comprise an annular lip that prevents plate **26** from being pulled out of housing **12** when the hook **18** embeds in the fish's mouth. Alternatively, plate **26** may be connected to spring **28** and spring **28** may be connected to housing **12**, to maintain a connection between the hook **18** and housing **12** (and to the attached fishing line) when the hook **18** is embedded in the fish's mouth.

[0021] According to another preferred embodiment, lure **10** also comprises a decorative hollow body **36** that is in a shape, such as a frog as shown in FIG. **6**, that would attract a fish to strike the lure. Other shapes for hollow body **36** may also be used. The hollow body **36** may also be part of the weedless guard system **16**, by protecting all or part of hook **18** from substantial contact with plant material when hook **18** is in a retracted position. Preferably, at least the barb end **34** of hook **18**, and more preferably all of hook **18** that is outside of housing **12** when in a retracted position, is disposed inside the hollow body **36**. Most preferably housing **12**, guard **20**, hook **18**, and brace **22** and barb **24** (if used) fit entirely within the hollow body **36** when the hook **18** is in a retracted position. The hollow body **36** preferably has one or more apertures **38** that correspond in size and location with hook(s) **18** that will allow hook(s) **18** to extend through the aperture(s) **38** in the hollow body **36** when guard **20** is disengaged by a fish strike and spring **28** pushes the hooks **18** rearwardly (in a direction substantially opposite from where the fishing line is attached to lure **10**). Alternatively, the barb end **34** of hook **18** may press up against an exterior surface of the hollow body when in a retracted position and other portions of lure **10** may be disposed inside the hollow body. A fishing line **40** may be attached to the hollow body (if used) or may pass through another aperture in the hollow body (such as a mouth of a frog body) to attach to attachment mechanism **14**.

[0022] According to another preferred embodiment, spring **28** may be integrally formed with a longitudinal portion of hook **18** without requiring use of plate **26**. According to another preferred embodiment, housing **12** may be integrally formed with attachment mechanism **14** and/or guard **20** or these parts may be connected together by adhesive, sonic welding, threaded connections, or a combination thereof. Housing **12** may also be integrally formed with hollow body **36** or housing and hollow body may be a single substantially cylindrical unit (particularly if a narrow bodied design, such as a worm, is used). Various materials, such as plastics, rubber, and metals, may be used for the parts of lure **10**, as will be understood by those of ordinary skill in the art. Lure **10** according to various preferred embodiments may be of differing sizes, depending on the type of fish that the user desires to catch. Typically lure **10** (without hollow body **36**) will preferably be around 2.5 to 5.5 inches long (in a retracted position) and around 1 to 2.5 inches wide. With hollow body **36**, lure **10** will preferably be around 2.75 to 7 inches long (in a retracted position) and around 1.25 to 4 inches wide. Other sizes may also be used depending on the size and type of fish desired. Lure **10** can be reused multiple times by simply pushing hook **18** inward toward housing **12** to compress spring **28** and re-engaging guard **20** with hook **18** (or barb **24** or other releasable connection used in weedless guard system **16**) to hold spring **28** in the retracted position in preparation for the next fish strike.

[0023] Although primarily described with a two hook configuration as shown in the figures, embodiments of the invention may use a single hook or any multiple hook configuration as will be understood by those of ordinary skill in the art. Those of ordinary skill in the art will also appreciate upon reading this specification and the description of preferred embodiments herein that modifications and alterations to the system may be made within the scope of the invention and it is intended that the scope of the invention disclosed herein be limited only by the broadest interpretation of the appended claims to which the inventors are legally entitled.

Claims

1. A fishing lure, comprising: a decorative hollow body comprising an aperture; a housing disposed inside and aligned longitudinally with the decorative hollow body; a hook extending at least partially from the housing, the hook comprising (1) a plate disposed inside the housing, (2) a longitudinal body coupled to the plate and extending from the plate longitudinally, (3) a curved portion extending from the longitudinal body and comprising a barb disposed at a distal end of the curved portion, wherein the hook is configured to be movable between a retracted position wherein

the barb of the hook is disposed within the decorative hollow body and an extended position wherein the barb of the hook extends through the aperture in the decorative hollow body and is disposed entirely outside of the decorative hollow body; a spring disposed inside the housing between and in contact with a forward end of the housing and the plate, the spring configured to selectively move the hook from the retracted position to the extended position in a longitudinal direction; a guard having a first end and a second end, wherein the first end of the guard is fixedly coupled to the housing, wherein the second end of the guard is releasably coupled to the hook, and wherein the guard is configured to selectively decouple from the hook and allow the spring to expand to move the hook from the retracted position to the extended position upon application of pressure to the guard.

2. The fishing lure of claim 1, wherein the housing comprises a substantially cylindrical body.

3. The fishing lure of claim 2, wherein the plate consists of a single plate, and wherein the housing comprises an annular lip extending inwardly from the substantially cylindrical body to retain the single plate within the housing.

4. The fishing lure of claim 3, further comprising: a sealing ring disposed around the plate.

5. The fishing lure of claim 3, wherein the plate forms a seal against the substantially cylindrical body of the housing.

6. The fishing lure of claim 1, wherein the hook comprises a second curved portion extending from the longitudinal body and comprising a second barb disposed at a distal end of the second curved portion.

7. The fishing lure of claim 6, further comprising: a brace disposed between the curved portion and the second curved portion.

8. The fishing lure of claim 7, wherein the second end of the guard is releasably coupled to the brace, and wherein the guard is configured to selectively decouple from the brace and allow the spring to expand to move the hook from the retracted position to the extended position upon application of pressure to the guard.

9. The fishing lure of claim 6, wherein the decorative hollow body comprises a second aperture through which the second barb and at least a portion of the second curved portion of the hook pass when the hook is moved from the retracted position to the extended position.

10. The fishing lure of claim 1, wherein the hook comprises a second longitudinal body coupled to the plate and extending from the plate longitudinally, and a second curved portion extending from the second longitudinal body and comprising a second barb disposed at a distal end of the second curved portion.

11. The fishing lure of claim 10, further comprising: a brace disposed between the curved portion and the second curved portion.

12. The fishing lure of claim 11, wherein the second end of the guard is releasably coupled to the brace, and wherein the guard is configured to selectively decouple from the brace and allow the spring to expand to move the hook from the retracted position to the extended position upon application of pressure to the guard.

13. The fishing lure of claim 10, wherein the decorative hollow body comprises a second aperture through which the second barb and at least a portion of the second curved portion of the hook pass when the hook is moved from the retracted position to the extended position.

14. The fishing lure of claim 1, wherein the guard engages with the curved portion of the hook to retain the hook in the retracted position.

15. The fishing lure of claim 1, wherein the guard does not engage with the barb of the hook to retain the hook in the retracted position.

16. The fishing lure of claim 1, wherein the decorative hollow body is shaped like a frog, a fish, a worm, or an insect.

17. A fishing lure comprising: a decorative hollow body comprising at least one aperture; a housing disposed inside and aligned longitudinally with the decorative hollow body; a hook assembly

extending at least partially from the housing, the hook assembly comprising (1) a plate disposed inside the housing, (2) at least one longitudinal body coupled to the plate and extending from the plate longitudinally, (3) a first curved portion extending from the at least one longitudinal body and comprising a first barb disposed at a distal end of the first curved portion, and (4) a second curved portion extending from the at least one longitudinal body and comprising a second barb disposed at a distal end of the second curved portion, wherein the hook assembly is configured to be movable between a retracted position wherein the first barb and the second barb of the hook assembly are disposed within the decorative hollow body and an extended position wherein the first barb and the second barb of the hook assembly extend through the at least one aperture in the decorative hollow body and are disposed entirely outside of the decorative hollow body; a spring disposed inside the housing between and in contact with a forward end of the housing and the plate, the spring configured to selectively move the hook assembly from the retracted position to the extended position in a longitudinal direction; a guard having a first end and a second end, wherein the first end of the guard is fixedly coupled to the housing, wherein the second end of the guard is releasably coupled to the hook assembly, and wherein the guard is configured to selectively decouple from the hook assembly and allow the spring to expand to move the hook assembly from the retracted position to the extended position upon application of pressure to the guard.

18. The fishing lure of claim 17, further comprising: a brace disposed between the first curved portion and the second curved portion, wherein the second end of the guard is releasably coupled to the brace, and wherein the guard is configured to selectively decouple from the brace and allow the spring to expand to move the hook assembly from the retracted position to the extended position upon application of pressure to the guard.

19. The fishing lure of claim 18, wherein at least one of the guard and the brace comprises a barb or a hook configured to releasably couple the guard and the brace.

20. The fishing lure of claim 17, wherein the decorative hollow body is shaped like a frog, a fish, a worm, or an insect.
